

Marty McFly has just perfected a wonderful new invention: time travel. This invention will allow him to travel back in time to two locations for just 5 minutes each before returning back to his current time. Wondering how he can make the most of this amazing opportunity, he decides to buy and sell stock. He will travel back in time to a low point in the stock market and purchase a number of shares. He will then travel to a time when the stock is at a high point and sell his shares. This is perfect! There is just one hitch: when should he buy and when should he sell?

Assignment Definition

You are given a file with stock prices. You are to search this file for the best time to buy and the best time to sell. The program will then display the amount of money made when 100 shares were bought and sold. Of course, we want this design to be as efficient as possible...

Please do the following:

- Create a representation of a solution to this problem using any combination of design tools presented this semester
- Evaluate the quality of your design using appropriate metrics presented this semester
- Validate the design by applying appropriate quality tools or techniques

Grading

	<i>Exceptional</i> 100%	<i>Good</i> 90%	<i>Acceptable</i> 70%	<i>Developing</i> 50%	<i>Missing</i> 0%
<i>Design Quality</i> 60%	The optimal solution was found	A workable solution was found that has acceptable performance characteristics	A correct solution was found with poor performance or a minor flaw exists with the presented solution	Part of the problem was solved but there exists serious issues with the design	The solution as presented is not on track to solve the problem
<i>Tools</i> 10%	The design was unambiguously and clearly described	An appropriate design tool was utilized and the tool was used without error	Minor ambiguity exists in the design presentation or a minor error exists in how the design tool was utilized	The design tool was utilized poorly	The solution does not demonstrate mastery of the design tools of the semester
<i>Metrics</i> 10%	Every nuance of the design was correctly characterized	A suitable metric was utilized and the metric was used without error	A minor error exists in the use of the metric	A serious problem exists in the use of the quality metric	No knowledge of a quality metric was demonstrated
<i>Quality</i> 20%	It is difficult to image how a defect could exist in the code	The design was verified through a suitable application of a quality technique	The design was not adequately verified, but knowledge of quality techniques were demonstrate	Familiarity with quality techniques were demonstrate somewhere in the submitted paper	No knowledge of quality techniques were demonstrated